

Program 1

```
# include < iostream >
# include < string >
using namespace std ;

class Student {
private :
int rollNumber ;
string name ;
float marks [3];
float total ;
float percentage ;
string result ;

public :
void acceptDetails () ;
cout << " Enter Roll Number : " ;
    ,
cin >> rollNumber ;
cin . ignore () ;
cout << " Enter Name : " ;
getline ( cin , name ) ;
cout << " Enter marks for 3
subjects : " ;
for (int i = 0; i < 3; i ++ ) {
cin >> marks [ i ] ;
}
}

void calculateResult () {
total = 0 ;
for (int i = 0; i < 3; i ++ ) {
total += marks [ i ] ;
}
}
```

```
percentage = total / 3.0;
```

```
if ( percentage >= 60) result = " First Class ";  
else if ( percentage >= 50) result = " Second Class ";  
else if ( percentage >= 40) result = " Pass ";  
else result = " Fail ";  
}
```

```
void displayDetails () {  
cout << "\n- - - - Student Details -----" << endl ;  
cout << " Roll Number : " << rollNumber << endl ;  
cout << " Name : " << name << endl ;  
cout << " Marks : ";  
for (int i = 0; i < 3; i ++ ) {  
4  
cout << marks [ i ] << " "cout  
}  
48  
cout << endl ;  
49  
cout << " Total Marks : " << total << endl ;  
50  
cout << " Percentage : " << percentage << "%" << endl ;  
51  
cout << " Result : " << result << endl ;  
52  
}  
53  
};  
54  
55  
int main () {  
56  
Student s ;  
57  
s . acceptDetails () ;  
58  
s . calculateResult () ;  
59  
s . displayDetails () ;  
60  
return 0;  
61  
}
```